

Benjamin Draves, Ph.D.

I'm a Staff Applied Scientist at Uber in NYC. I build recommendation systems and ML targeting systems to grow Uber's consumer and earner businesses. I hold a Ph.D. in Statistics from Boston University where I studied graph embedding methods.

Experience

Staff Applied Scientist – Uber **2022–Present**

New York City, NY

- [Developed recommendation systems](#) to recommend restaurants, offers, and grocery and retail stores to grow Uber's delivery business.
- Created an [ML targeting system](#) to re-engage churned earners with incentives, driving efficient supply growth in all-markets, globally.
- Organized a weekly Learning and Development series and a CausalML reading group for [150-person DS org](#).

Recommendation Systems

- Built recommendation models...
 - for restaurants/offers utilizing [learn to rank XGBoost](#) models
 - for grocery/retail stores using [relational collaborative filtering](#) with [PySpark](#)
- Built models to determine...
 - user's most likely next delivery location using [XGBoost](#)
 - user's cuisine preferences using [graph embedding](#) with [PySpark](#)

ML Targeting Systems

- Built targeting models to predict..
 - an earner's responsiveness to an incentive using [CausalML meta-learners \(S, X, T, R\)](#)
 - a rider's preferred mobility product using [multi-label classification models](#) in [Keras](#)
- Built a budget pacing system to allocate incentive spend across time/geos using [PID controllers](#) and [convex optimization](#)

Data Scientist Intern – HP **2021**

Remote – Boston, MA

- Designed and implemented an experimentation platform utilizing [causal inference](#) methods to assess changes in global pricing strategies.
- Deployed a [gradient-boosted regression model](#) to a production pricing tool.

Lead Statistical Consultant - Boston University **2019-2020**

Boston, MA

- Oversaw team of 14 Masters students working in BU's consulting center.
- Provided consulting services to 35 clients in industry and academia.

Education

Ph.D., Master of Arts: Statistics **2017-2022**

Boston University - Boston, MA

- [Dissertation](#): Joint Spectral Embeddings of Random Dot Product Graphs
- Research Areas: Machine Learning, Matrix Analysis, Multivariate Stat.
- GPA: 3.98. Qualifying exams: Theoretical Probability, Applied Statistics

Bachelor of Science: Mathematics **2014 - 2017**

Lafayette College - Easton, PA

- GPA: 3.90. Graduated summa cum laude with honors and thesis.

Contact

Phone

(330) 428-5025

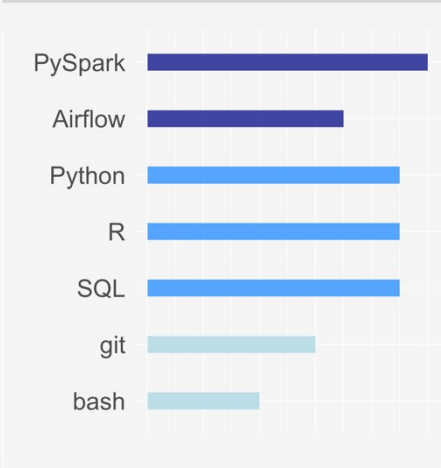
E-mail

benjamin.draves@gmail.com

Website

dravesb.github.io

Languages



Leadership & Awards

- Invited Talks – UChicago Data Science Institute, Lafayette College
- Committee Chair – [BU Data Science Mentoring Circles](#)
- BU Excellence in Student Teaching Award
- BU Student Chapter of the ASA - Board Member
- Barge Oratorical Prize – most compelling thesis defense
- Mitman Math Award – most outstanding math major

Hobbies & Interests

